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Video Streaming Thinking: VideoLLMs Can Watch and Think Simultaneously

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Abstract. Online Video Large Language Models (VideoLLMs) play a critical role in supporting responsive, real-time interaction. Existing methods focus on streaming perception, lacking a synchronized logical reasoning stream. However, directly applying test-time scaling methods incurs unacceptable response latency. To address this trade-off, we propose ***V**ideo **S**treaming **T**hinking* (VST), a novel paradigm for streaming video understanding. It supports a *thinking-while-watching* mechanism, which activates reasoning over incoming video clips during streaming. This design improves timely comprehension and coherent cognition while preserving real-time responsiveness by amortizing LLM reasoning latency over video playback. Furthermore, we introduce a comprehensive post-training pipeline that integrates VST-SFT, which structurally adapts the offline VideoLLM to causal streaming reasoning, and VST-RL, which provides end-to-end improvement through self-exploration in a multi-turn video interaction environment. Additionally, we devise an automated training-data synthesis pipeline that uses video knowledge graphs to generate high-quality streaming QA pairs, with an entity–relation grounded streaming Chain-of-Thought to enforce multi-evidence reasoning and sustained attention to the video stream. Extensive evaluations show that VST-7B performs strongly on online benchmarks, *e.g.* 79.5% on StreamingBench and 59.3% on OVO-Bench. Meanwhile, VST remains competitive on offline long-form or reasoning benchmarks. Compared with Video-R1, VST responds 15.7× faster and achieves +5.4% improvement on VideoHolmes, demonstrating higher efficiency and strong generalization across diverse video understanding tasks. Code, data, and models have been released at `https://github.com/1ranGuan/VST`.

Keywords: Streaming Video Understanding · CoT · VideoLLM

1 Introduction

Online video understanding enables Video Large Language Models (VideoLLMs) to interpret streaming visual inputs and respond in real time, making it particularly valuable for embodied intelligence [9,13] and interactive AI assistants [3,8].

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視頻串流思考：視頻大語言模型能同時觀看與思考

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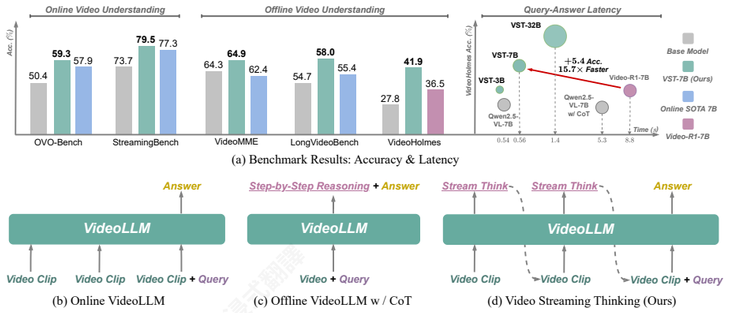
摘要。線上視頻大語言模型（視頻大語言模型）在支援響應式、實時互動方面扮演關鍵角色。現有方法主要著重於串流感知，缺乏同步的邏輯推理串流。然而，直接應用測試時刻縮放方法會導致不可接受的響應延遲。為了解決這種取捨關係，我們提出視頻串流思考（VST），一種新的串流視頻理解範式。它支援邊看邊思考機制，在串流過程中激活對即時到來的影片片段的推理。這種設計通過將大語言模型推理延遲分攤到影片播放過程中，提升了及時理解與連貫認知，同時保證了實時響應性。此外，我們引入了一套全面的後訓練流程，集成了VST-SFT，該技術結構性地將離線視頻大語言模型適應到因果串流推理，以及VST-RL，它通過在多輪影片互動環境中的自我探索實現端到端的改進。此外，我們設計了一條自動化的訓練數據合成流程，該流程使用視頻知識圖生成高質量的串流問答對，並使用實體-關係接地串流思考鏈來強制多證據推理並保持對影片串流的持續關注。廣泛的評估表明，VST-7B 在線上基準上表現出色，例如在StreamingBench上達到79.5%，在OVO-Bench上達到59.3%。同時，VST 在離線長文本或推理基準上仍然具有競爭力。與Video-R1相比，VST響應 15.7× 更快，在VideoHolmes上實現 +5.4%的改進，展現了更高的效率以及在不同視頻理解任務上的強泛化能力。代碼、數據和模型已釋放在https://github.com/1ranGuan/VST。

關鍵字：串流視頻理解 · 因果推理 · VideoLLM

1 緒論

線上影片理解讓視頻大語言模型（VideoLLMs）能夠解釋串流視覺輸入並實時回應，使其特別適用於
對具身智能 [9,13] 與互動AI助理 [3,8] 而言，這是相當有價值的。

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(a) Benchmark Results: Accuracy & Latency

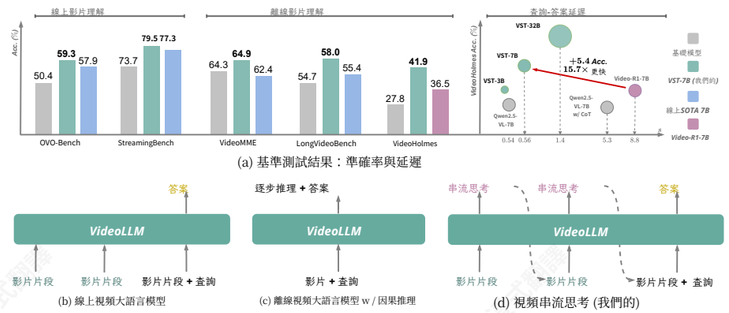
(b) Online VideoLLM (c) Offline VideoLLM w / CoT (d) Video Streaming Thinking (Ours)

Fig. 1: Benchmark results and paradigm comparison. (a) VST-7B delivers strong performance on online and offline video understanding benchmarks while maintaining low QA latency. (b) Existing streaming VideoLLMs focus on efficient streaming processing, but lack explicit analytical reasoning. (c) VideoLLM with CoT performs heavy post-query step-by-step reasoning to improve performance, but incurs high QA latency. (d) Our method introduces proactive pre-query reasoning, interleaving it with video consumption to achieve both strong performance and efficient responsiveness.

Unlike offline methods that benefit from post-hoc global access to the entire video [1, 22, 24, 28, 54], the core challenges of online video understanding lie in strict temporal causality, real-time processing, and a finite context window.

Several prior methods have been proposed to address the challenges of online video understanding. As shown in Fig. 1(b), they primarily improve context-window efficiency by explicitly managing visual tokens for compression [41, 45, 58, 62] or by retrieving from the KV cache [6, 34, 57]. However, these methods primarily focus on streaming perception and treat the management of visual features as a form of memory, with limited involvement of the LLM itself and no explicit reasoning or analytical deliberation. To fill this missing piece, one promising direction inspired by offline video understanding is to apply test-time scaling via Chain-of-Thought (CoT) to elicit stronger reasoning ability [4, 10, 14, 15, 29, 52, 63, 69], as shown in Fig. 1(c). Nevertheless, directly performing step-by-step reasoning after the user query can significantly increase QA response latency, making it difficult to meet strict real-time requirements in online scenarios.

In this paper, we introduce the ***V**ideo **S**treaming **T**hinking* (VST) to resolve the trade-off between explicit reasoning and real-time responsiveness, shifting the LLM backend from passive waiting to active, intermittent reasoning during video consumption. This design is inspired by insights from human cognition. Findings on *neural coupling* [19,42] suggest that the logical flow in the brain synchronizes closely with the influx of external information, fostering the perception of current signals and their synthesis into a coherent understanding. Similarly, as illustrated in Fig. 1(d), our method continuously processes incoming video clips and produces intermediate thoughts in real time. This eliminates the need to



(a) 基準測試結果：準確率與延遲

(b) 線上視頻大語言模型 (c) 離線視頻大語言模型 w / 因果推理 (d) 視頻串流思考 (我們的)

圖 1：基準結果與範式比較。(a) VST-7B 在線上和離線視頻理解基準上表現強勢，同時保持低問答延遲。(b) 現有串流視頻大語言模型主要關注高效能串流處理，但缺乏明確的分析推理。(c) 具備思維鏈的視頻大語言模型進行重點的查詢後步驟式推理以提升表現，但會導致高問答延遲。(d) 我們的方法引進主動的查詢前推理，與視頻消耗交錯進行，以達到強勁的表現和高效的響應。

與可透過事後處理獲得整個影片 [1, 22, 24, 28, 54], 全局資訊的離線方法不同，線上影片理解的核心理戰挑戰在於嚴格的時間因果關係、實時處理，以及有限的上下文視窗。

先前已有數種方法被提出來應對線上視頻理解的挑戰。如圖 1(b) 所示，它們主要透過明確管理視覺標記進行壓縮 [41, 45,58, 62] 或從 KV 缓存中檢索 [6, 34, 57] 來提升上下文窗口效率。然而，這些方法主要關注串流感知，將視覺特徵的管理視為一種記憶形式，對大語言模型的參與有限，且無明確的推理或分析判斷。為了填補這個缺失環節，一個受離線視頻理解啟發的有希望的方向是透過思維鏈 (CoT) 在測試時刻縮放來引導更強的推理能力 [4, 10, 14, 15, 29, 52, 63, 69], ，如圖 1(c) 所示。然而，直接在用戶查詢後進行步驟式推理會顯著增加問答響應延遲，使其難以滿足線上場景嚴格的實時要求。

在本篇論文中，我們介紹了視頻串流思考 (VST)，以解決明確推理與實時響應之間的取捨，將大語言模型後端從被動等待轉變為在視頻消費過程中主動、間歇的推理。此設計受人类认知的啟示所鼓舞。神經耦合 [19,42] 的研究發現表明，大腦中的邏輯流程與外部信息的流入緊密同步，促進了當前信號的感知及其綜合成一致的理解。類似地，如圖 1(d) 所示，我們的方法持續處理輸入的影片片段，並實時產生中間思考。這消除了需要

defer heavy computation until the query arrives, which is a common limitation of offline VideoLLMs with CoT [4, 10, 48]. This *thinking-while-watching* mechanism maintains a coherent internal state over the stream, ensuring that the final response is grounded in a deeply processed understanding of the historical context. By front-loading and amortizing the reasoning cost ahead of query arrival, VST preserves the low QA latency required in streaming scenarios.

We instantiate this paradigm with a dedicated post-training pipeline that combines supervised fine-tuning (VST-SFT) and reinforcement learning (VST-RL). Concretely, we cast streaming thinking as a multi-turn conversation, where the model incrementally writes textual thoughts to an external memory while observing incoming video clips under a constrained visual context window. In the VST-SFT stage, we align the model with the desired streaming reasoning protocol by learning from off-policy demonstrations that strictly respect temporal causality, thereby bootstrapping its basic *thinking-while-watching* capability. Building upon this initialization, the VST-RL stage performs end-to-end reinforcement learning with verifiable rewards, encouraging the model to make intermediate reasoning steps that improve downstream question answering under realistic online conditions.

Due to the scarcity of existing data for video streaming thinking, we develop an automated synthesis pipeline to support our training, particularly the VST-SFT stage that requires high-quality reasoning demonstrations. Specifically, we model entities and their temporal relationships within long videos as knowledge graphs. By sampling paths from these graphs to form evidence chains, we prompt an offline VideoLLM to generate complex QA pairs and their corresponding intermediate CoTs. This design enforces multi-hop reasoning across diverse visual evidence while ensuring strict alignment between the generated thoughts and the video context. Ultimately, we synthesize a large-scale dataset comprising 100K high-quality streaming reasoning samples.

We conducted extensive evaluations across multiple online and offline video understanding benchmarks (see Fig. 1(a)). The results show that our method achieves state-of-the-art performance compared to existing online VideoLLMs, while remaining competitive on offline video understanding benchmarks. Notably, VST performs particularly well on long-form videos that require comprehensive plot comprehension and multi-step reasoning. Moreover, compared to Video-R1, our method achieves higher accuracy while significantly reducing QA latency, demonstrating that VST is a viable test-time scaling approach that meets the requirements of streaming scenarios.

In summary, our main contributions are as follows:

- We propose the VST paradigm to interleave active explicit CoT generation with continuous video streams, enabling amortized test-time scaling with real-time responsiveness.
- A knowledge-graph-based data synthesis pipeline and a dedicated post-training recipe (VST-SFT and VST-RL) are introduced to adapt an offline VideoLLM to streaming settings with strong streaming reasoning capabilities.

延遲重複計算直到查詢到來，這是具有因果推理 [4, 10, 48] 的離線視頻大語言模型的一個常見限制。這種邊看邊思考的機制在串流中維持一個連貫的內部狀態，確保最終的回應是基於對歷史背景的深度處理理解。透過在查詢到來之前預加負擔並分攤推理成本，VST 保留了串流場景中所需的低問答延遲。

我們藉由一個專用的訓練後管線來實例化這個典範，該管線結合了監督微調（VST-SFT）和強化學習（VST-RL）。具體來說，我們將串流思考視為一個多輪對話，模型在觀察到來自受限制視覺上下文的影片片段時，會逐步將文字思考寫入外部記憶體。在VST-SFT階段，我們透過學習嚴格遵守時間因果性的離線示範，來將模型與期望的串流推理協議對齊，從而啟動其基本的邊看邊思考能力。在這個初始化基礎上，VST-RL階段進行端到端的強化學習，並使用可驗證的回饋，鼓勵模型在真實的線上條件下進行能夠改善下游問答的 intermediate reasoning steps。

由於現有數據對於視頻串流思考的稀缺性，我們開發了一個自動合成管線來支持我們的訓練，特別是VST-SFT階段，該階段需要高品質的推理示範。具體來說，我們將長影片內實體及其時間關係建模為知識圖。透過從這些圖中採樣路徑來形成證據鏈，我們提示一個離線視頻大語言模型（VideoLLM）來生成複雜的QA對及其對應的 intermediate CoTs。這種設計強制在多樣化的視覺證據之間進行多跳推理，同時確保生成的思考與影片上下文之間嚴格對齊。最終，我們合成了一個包含100K高品質串流推理樣本的規模龐大的數據集。

我們在多個線上和離線影片理解基準（見圖 1(a)）上進行了廣泛評估。結果顯示，與現有的線上 VideoLLMs 相比，我們的方法實現了最優性能，同時在離線影片理解基準上保持競爭力。值得注意的是，VST 在需要全面劇情理解和多步推理的長文本視頻上表現特別出色。此外，與 Video-R1 相比，我們的方法實現了更高的準確性，同時顯著降低了問答延遲，證明 VST 是一種可行的測試時刻縮放方法，滿足串流場景的要求。

總結來說，我們的主要貢獻如下：

- 我們提出VST典範，將主動明確CoT生成與連續影片串流交錯，實現分攤測試時間縮放與實時反應能力。
- 一個基於知識圖表的数据合成管線，以及一個專用的訓練後配方（VST-SFT 和 VST-RL）被介紹，用於將離線視頻大語言模型適應到具有強強串流推理能力的串流設定。

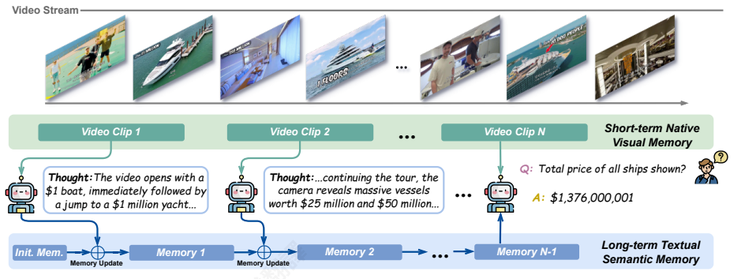


Fig. 2: Illustration of the Video Streaming Thinking pipeline. The model employs a streaming thought mechanism to compress visual dynamics into a long-term textual memory. Combined with the short-term visual buffer, this enables efficient reasoning over indefinite video streams with fixed memory budgets.

- Extensive evaluations across multiple online and offline video understanding benchmarks demonstrate state-of-the-art performance. In addition, compared to offline CoT VideoLLM, our method provides significantly lower QA latency.

2 Method

2.1 The *Video Streaming Thinking* (VST) Paradigm

We formulate VST as a multi-round video conversation task operating within a constrained context window, as illustrated in Fig. 2. Unlike previous online VideoLLMs, our model leverages streaming intervals before a user query to proactively reason about the content via autoregressive textual generation. This process synthesizes key visual details and event dynamics into a dual-memory system: maintaining a *short-term native video memory* for the current visual context, while accumulating a *long-term textual semantic memory* of past events.

Formally, given a video stream $\mathcal{V}$, let $\mathbf{v}_i$ denote the visual features for the i-th frame. We accumulate these incoming features into discrete clips $\mathbf{c}^k = \{\mathbf{v}_i\}_{i=\tau_{k-1}+1}^{\tau_k}$, where the boundary τ_k is set when the accumulated visual tokens reach the preset capacity L. At each interval k, conditioned on the current clip $\mathbf{c}^k$ and the accumulated memory $\mathbf{m}^{k-1}$, the LLM generates a streaming thought $\mathbf{z}^k$ by sampling from the distribution $\mathbf{z}^k \sim p(\mathbf{z} \mid \mathbf{c}^k, \mathbf{m}^{k-1})$. Here, $\mathbf{z}^k$ summarizes the essential semantics of the current video segment, preserving the continuity of the overall thought process. For the long-term textual memory, we employ a memory update function $\mathbf{m}^k = \text{Update}(\mathbf{m}^{k-1}, \mathbf{z}^k)$, which adopts a simple first-in-first-out strategy to evict the earliest memory entries.

This iterative reasoning process continues until step K, when a user query $\mathbf{q}$ is received. Upon this trigger, the LLM generates the final response $\mathbf{y}$ based on the accumulated thoughts $\{\mathbf{z}^k\}_{k=1}^{K-1}$ and the latest visual context $\mathbf{c}^K$. Consequently,

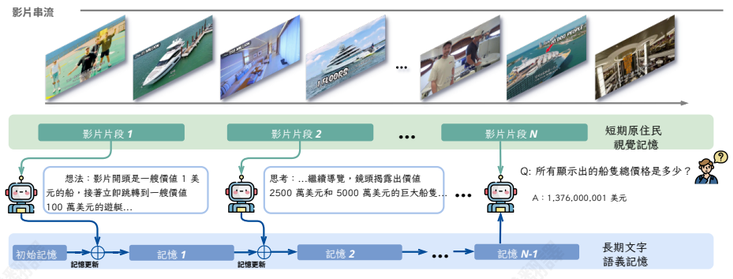


圖 2：視頻串流思考 (VST) 的流程圖解。該模型採用串流思考機制，將視覺動態壓縮為長期文字記憶。結合短期視覺緩衝區，這使得模型能在固定記憶預算下，對無限期的視頻串流進行高效推理。

- 在多個線上和離線影片理解基準上的廣泛評估顯示出最優性能。此外，與離線因果推理 VideoLLM 相比，我們的方法提供了顯著更低的問答延遲。

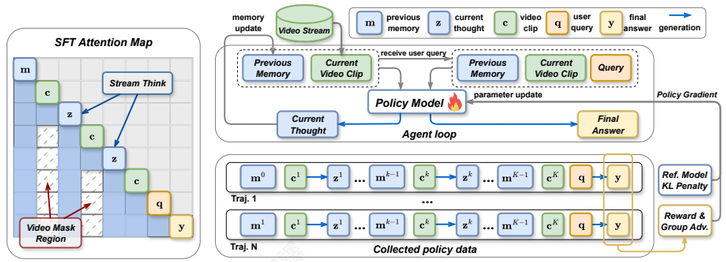
2 方法

2.1 視頻串流思考 ***(VST)*** 范式

我們將 VST 定義為一個在受限上下文窗內進行的多輪視頻對話任務，如圖 2 所示。與之前的線上視頻大語言模型 (VideoLLMs) 不同，我們的模型利用用戶查詢前的串流間隔，透過自回歸文字生成來主動推理解釋內容。此過程將關鍵視覺細節和事件動態合成為雙記憶系統：維持一個短期原生視頻記憶以處理當前視覺語境，同時積累過去事件的長期文字語義記憶。

形式上，給定一個視頻串流 $\mathcal{V}$，令 $\mathbf{v}_i$ 表示第 i 幀的視覺特徵。我們將這些輸入特徵積累為離散片段 $\mathbf{c}^k = \{\mathbf{v}_i\}_{i=\tau_{k-1}+1}^{\tau_k}$，當累積的視覺 token 到達預設容量 L 時，邊界 τ_k 便會被設定。在每個間隔 k 中，基於當前影片 $\mathbf{c}^k$ 和累積的記憶 $\mathbf{m}^{k-1}$，LLM 透過從分佈 $\mathbf{z}^k \sim p(\mathbf{z} \mid \mathbf{c}^k, \mathbf{m}^{k-1})$ 中採樣來生成串流思考 $\mathbf{z}^k$。這裡，$\mathbf{z}^k$ 總結了當前影片片段的關鍵語義，保留了整體思考過程的連續性。對於長期文字記憶，我們採用一個記憶更新函數 $\mathbf{m}^k = \text{Update}(\mathbf{m}^{k-1}, \mathbf{z}^k)$，該函數採用簡單的先進先出策略來移除最早期的記憶條目。

這個迭代推理過程會持續進行，直到步驟 K，當使用者查詢 $\mathbf{q}$ 被接收時。在此觸發下，大語言模型會根據累積的想法 $\{\mathbf{z}^k\}_{k=1}^{K-1}$ 與最新的視覺語境 $\mathbf{c}^K$ 生成最終回應 $\mathbf{y}$。因此，



(a) VST-SFT (b) VST-RL

Fig. 3: Overview of the training pipeline. (a) VST-SFT applies a streaming attention mask to enforce temporal causality, restricting attention to the current visual buffer and history textual context. (b) VST-RL performs on-policy optimization via an agentic loop, improving the quality of streaming thoughts through verifiable rewards computed solely from the final answer.

the joint probability of the thoughts and the final answer is decomposed as:

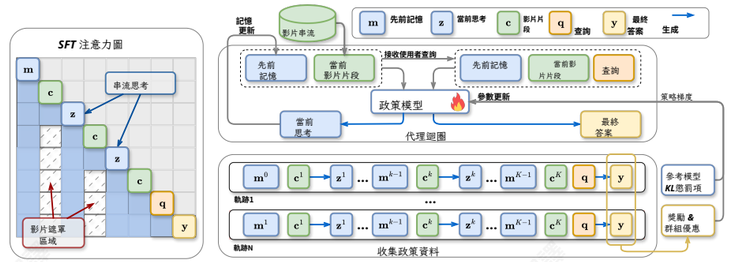
$$p(\mathbf{y}, \{\mathbf{z}^k\}_{k=1}^{K-1} \mid \mathbf{q}, \mathcal{V}) = \underbrace{p(\mathbf{y} \mid \mathbf{q}, \mathbf{c}^K, \mathbf{m}^{K-1})}_{\text{Direct Answer}} \prod_{k=1}^{K-1} \underbrace{p(\mathbf{z}^k \mid \mathbf{c}^k, \mathbf{m}^{k-1})}_{\text{Streaming Thinking}}. \quad (1)$$

This formulation yields two distinct advantages. 1) It amortizes the computational cost of Chain-of-Thought (CoT) generation over the pre-query phase. This strategy effectively achieves test-time scaling to boost performance without incurring additional latency at the moment of user interaction. 2) The sequential generation of thoughts naturally aligns with the temporal causality inherent in streaming videos. This structure facilitates the adaptation of offline models to online scenarios by mirroring the progressive nature of the video stream.

2.2 Training Method for VST

To instantiate the VST paradigm introduced in Sec. 2.1, we develop a two-stage post-training pipeline that combines supervised fine-tuning (VST-SFT) and reinforcement learning (VST-RL), progressively endowing an offline VideoLLM with streaming thinking capabilities. The VST-SFT stage adapts the offline model to the temporal causality of streaming video, while learning reasoning capabilities from off-policy expert data. Subsequently, VST-RL transitions the model from off-policy imitation to on-policy RL, and refines these learned capabilities for further end-to-end improvement.

Stage 1: VST-SFT. We initiate the training pipeline with SFT to instill the streaming thought mechanism into the offline VideoLLM. For a training instance,



(a) VST-SFT (b) VST-RL

圖3：訓練流程概述。(a) VST-SFT 應用串流注意力遮罩以強制時間因果關係，將注意力限制在當前視覺緩衝區和歷史文本上下文。(b) VST-RL 透過代理迴圈進行策略優化，透過僅從最終答案計算的可驗證回饋來提高串流思考的品質。

想法與最終答案的聯合機率被分解為：

$$p(\mathbf{y}, \{\mathbf{z}^k\}_{k=1}^{K-1} \mid \mathbf{q}, \mathcal{V}) = \underbrace{p(\mathbf{y} \mid \mathbf{q}, \mathbf{c}^K, \mathbf{m}^{K-1})}_{\text{Direct Answer}} \prod_{k=1}^{K-1} \underbrace{p(\mathbf{z}^k \mid \mathbf{c}^k, \mathbf{m}^{k-1})}_{\text{Streaming Thinking}}. \quad (1)$$

這種公式產生兩個獨特的優勢。1) 它將思維鏈 (CoT) 生成計算成本分攤到查詢前階段。這種策略有效地實現測試時刻縮放，以提升性能，而無需在用戶互動時刻引入額外延遲。2) 思考的序列生成自然地與串流影片中內在的時間因果關係一致。這種結構促進了離線模型對線上場景的適應，透過模擬影片串流的逐步性質。

2.2 VST 的訓練方法

為了實現 Sec. 2.1 中介紹的 VST 范式，我們開發了一個兩階段後訓練流程，結合監督微調（VST-SFT）和強化學習（VST-RL），逐步賦予離線 VideoLLM 串流思考能力。VST-SFT 階段使離線模型適應串流影片的時間因果關係，同時從離策專家數據中學習推理能力。接下來，VST-RL 將模型從離策模擬轉變為在策強化學習，並精煉這些學習到的能力以進行進一步的端到端改進。

第一階段：VST-SFT。我們以 SFT 啟動訓練流程，將串流思考機制植入離線視頻大語言模型。對於一個訓練實例，

we explicitly formulate the sequence as:

$$\mathcal{S} = \Big(\mathbf{m}^0, (\mathbf{c}^1, \mathbf{z}^1), \ldots, (\mathbf{c}^{K-1}, \mathbf{z}^{K-1}), \mathbf{c}^K, \mathbf{q}, \mathbf{y}\Big). \tag{2}$$

Here, $\mathbf{m}^0$ denotes the initial memory, and $(\mathbf{c}^k, \mathbf{z}^k)$ represent the interleaved video clips and streaming thoughts. The sequence concludes with the final clip $\mathbf{c}^K$, user query $\mathbf{q}$, and ground truth response $\mathbf{y}$.

To align with the streaming inference architecture, we apply a streaming video attention mask. As depicted in Fig. 3(a), this mask restricts the model's attention to a fixed-size window of recent visual tokens, mirroring the short-term visual buffer used during inference. Specifically, let M be the additive attention mask. Let $\mathbb{I}_v(j) \in \{0, 1\}$ indicate whether the j-th token is a visual token, and let L denote the visual buffer size. Therefore, the attention mask can be written as:

$$M_{i,j} = \begin{cases} 0, & j \le i \text{ and } \Big(\mathbb{I}_v(j) = 0 \text{ or } \sum_{t=j+1}^{i} \mathbb{I}_v(t) < L\Big) \\ -\infty, & \text{otherwise} \end{cases} \tag{3}$$

In this way, the model can only access a sliding window of the latest L visual tokens, while all non-visual tokens remain fully visible under the causal constraint. Furthermore, to accommodate context length constraints while handling long-form videos, we implement a temporal segmentation strategy. The original sequence $\mathcal{S}$ is sliced into consecutive segments $\{\mathbf{s}_n\}_{n=1}^{M}$, defined as:

$$\mathbf{s}_n = \begin{cases} \Big(\mathbf{m}^{n-1}, \{(\mathbf{c}^k, \mathbf{z}^k)\}_{k=T_{n-1}+1}^{T_n}\Big), & n < M \\ \Big(\mathbf{m}^{n-1}, \{(\mathbf{c}^k, \mathbf{z}^k)\}_{k=T_{n-1}+1}^{K-1}, \mathbf{c}^K, \mathbf{q}, \mathbf{y}\Big), & n = M \end{cases} \tag{4}$$

where T_n denotes the cut-off index for the n-th segment. The memory state is updated recursively across segments following $\mathbf{m}^n = \text{Update}(\mathbf{m}^{n-1}, \{\mathbf{z}^k\}_{k=T_{n-1}+1}^{T_n})$. During SFT, we apply the standard next-token prediction loss exclusively to the streaming thoughts $\{\mathbf{z}^k\}_{k=1}^{K-1}$ and the final response $\mathbf{y}$, treating visual tokens and historical memory as conditioning inputs.

Stage 2: VST-RL. Building upon the supervised foundation, we introduce VST-RL to transition the model from off-policy imitation to on-policy self-improvement. The RL training process consists of two main phases: trajectory rollout and policy gradient optimization.

As shown in the upper part of Fig. 3(b), the rollout phase operates as an agentic loop. The policy model interacts with the streaming environment to generate a trajectory $\mathcal{T}$ following the predefined joint probability in Eq. (1), where the streaming thoughts $\hat{\mathbf{z}}^k$ and the final response $\hat{\mathbf{y}}$ are sequentially sampled from the sampling policy $\pi_{\theta'}$. After collecting a group of N trajectories $\{\mathcal{T}_i\}_{i=1}^{N}$, we employ a GRPO [15, 33, 60, 61] strategy to optimize the policy model. We compute the reward $\mathbf{r_i}$ solely based on the final answer $\mathbf{y_i}$ via verifiable reward functions. To encourage the model to generate useful streaming thoughts, the

我們明確制定序列為：

$$\mathcal{S} = \Big(\mathbf{m}^0, (\mathbf{c}^1, \mathbf{z}^1), \ldots, (\mathbf{c}^{K-1}, \mathbf{z}^{K-1}), \mathbf{c}^K, \mathbf{q}, \mathbf{y}\Big). \tag{2}$$

在此，m^0 代表初始記憶，而 $(\mathbf{c}^k, \mathbf{z}^k)$ 代表交錯的影片片段與串流思考。序列結尾為最後的片段 $\mathbf{c}^K$、使用者查詢 q，以及真實回應 y。

為與串流推論架構對齊，我們應用一個串流影片注意力遮罩。如圖 3(a) 所示，此遮罩限制模型的注意力在固定大小的近期視覺 token 窗口中，反映推論期間使用的短期視覺緩衝區。具體來說，讓 M 為加性注意力遮罩。讓 $\mathbb{I}_v(j) \in \{0, 1\}$ 指示第 j 個 token 是否為視覺 token，讓 L 表示視覺緩衝區大小。因此，注意力遮罩可以寫為：

$$M_{i,j} = \begin{cases} 0, & j \le i \text{ and } \Big(\mathbb{I}_v(j) = 0 \text{ or } \sum_{t=j+1}^{i} \mathbb{I}_v(t) < L\Big) \\ -\infty, & \text{otherwise} \end{cases} \tag{3}$$

這樣，模型只能存取最新 L視覺 token 的滑動窗口，而所有非視覺 token 在因果約束下保持完全可見。此外，為了在處理長文本影片時兼顧上下文長度限制，我們實施了一種時間分割策略。原始序列 $\mathcal{S}$ 被切分成連續的片段 $\{\mathbf{s}_n\}_{n=1}^{M}$，定義如下：

$$\mathbf{s}_n = \begin{cases} \Big(\mathbf{m}^{n-1}, \{(\mathbf{c}^k, \mathbf{z}^k)\}_{k=T_{n-1}+1}^{T_n}\Big), & n < M \\ \Big(\mathbf{m}^{n-1}, \{(\mathbf{c}^k, \mathbf{z}^k)\}_{k=T_{n-1}+1}^{K-1}, \mathbf{c}^K, \mathbf{q}, \mathbf{y}\Big), & n = M \end{cases} \tag{4}$$

T_n 表示第 n個片段的切斷索引。記憶狀態會沿著遵循 $\mathbf{m}^n = \text{Update}(\mathbf{m}^{n-1}, \{\mathbf{z}^k\}_{k=T_{n-1}+1}^{T_n})$ 的片段遞迴更新。在 SFT 模式下，我們僅將標準的下一個詞元預測損失應用於串流思考 $\{\mathbf{z}^k\}_{k=1}^{K-1}$ 和最終回應 y，將視覺詞元和歷史記憶視為條件輸入。

第二階段：VST-RL。在監督基礎上，我們引入 VST-RL 來將模型從離策模仿過渡到自策改進。RL 訓練過程包含兩個主要階段：軌跡擴展和政策梯度優化。

如圖3(b)上方所示，展開階段運作為一個代理迴圈。政策模型與串流環境互動，以生成一條符合公式(1)中預設聯合機率的軌跡 $\mathcal{T}$，其中串流思考 $\hat{\mathbf{z}}^k$ 與最終回應ˆy是依序從採樣策略 $\pi_{\theta'}$中採樣的。在收集了一組 N 軌跡 $\{\mathcal{T}_i\}_{i=1}^{N}$,後，我們採用GRPO [15, 33, 60, 61] 策略來優化政策模型。我們僅根據最終答案$\mathrm{y_i}$，透過可驗證的回報函數來計算回報$\mathrm{r_i}$。為了鼓勵模型生成有用的串流思考，我們將計算出的優勢分配給整個軌跡 $\mathcal{T}$中所有生成的詞元。政策梯度目標計算如下：

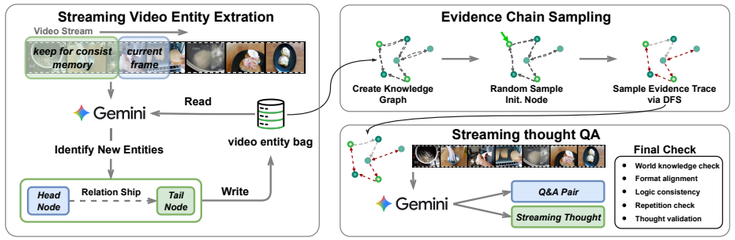


Fig. 4: Stream-Thought QA data curation pipeline. We incrementally extract video entities and relations to build a knowledge graph, sample multi-hop evidence chains, and use Gemini to generate streaming QA pairs with grounded streaming thoughts, followed by automatic filtering.

calculated advantage is assigned to all generated tokens within the entire trajectory $\mathcal{T}_i$. The policy gradient objective is calculated as:

$$\mathcal{J}_{\mathrm{RL}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\{\mathcal{T}_i\}_{i=1}^{N}\sim\pi_{\theta'}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^{N}|\mathcal{T}_i|} \sum_{i=1}^{N}\sum_{t=1}^{|\mathcal{T}_i|} \left(\mathcal{L}_{i,t}^{\mathrm{clip}}(\theta) - \beta D_{\mathrm{KL}}(\pi_\theta||\pi_{\mathrm{ref}}) \right) \right], \quad (5)$$

$$\mathcal{L}_{i,t}^{\mathrm{clip}}(\theta) = \min \left[\gamma_t(\theta)\hat{A}_i, \mathrm{clip}\left(\gamma_t(\theta), 1-\epsilon_{\mathrm{low}}, 1+\epsilon_{\mathrm{high}}\right)\hat{A}_i \right]. \quad (6)$$

Where $|\mathcal{T}_i|$ denotes the total number of generated tokens in trajectory $\mathcal{T}_i$, $\gamma_t(\theta)$ represents the probability ratio between π_θ and the sampling policy $\pi_{\theta'}$ at step t, $\hat{A}_i = r_i - \mathrm{mean}(R)$ is the group relative advantage, and $\epsilon_{\mathrm{low}}, \epsilon_{\mathrm{high}}$ are the clipping hyperparameters follow DAPO [61].

2.3 Data Synthesis Pipeline for VST

We generate a set of video streaming thought data to support VST training, motivated by the fact that most existing chain-of-thought (CoT) datasets target offline VideoLLMs with a global, hindsight view of the entire video, making it difficult to avoid information leakage under causal streaming constraints. To this end, we introduce an automated data generation pipeline grounded in knowledge graphs. As illustrated in Fig. 4, the pipeline produces high-quality training examples with explicit reasoning paths through streaming video entity extraction, evidence chain sampling, and streaming thought QA synthesis.

Streaming Video Entity Extraction. To build a temporally consistent knowledge graph, we maintain an *entity bank* and extract triples from a sliding window over the video stream. We segment the video into N scene clips with PySceneDetect. For each incoming clip, an offline VideoLLM (*e.g.* Gemini 3.0 flash) updates the entity bank by adding newly observed entities and relations

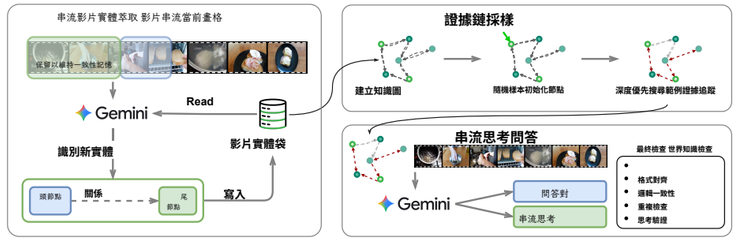


圖4：串流-思考問答數據策劃流程。我們逐步提取視頻實體和關係來建立知識圖，採樣多跳證據鏈，並使用Gemini生成有基礎的串流思考的串流問答對，接著進行自動篩選。

計算出的優勢被分配給軌跡 $\mathcal{T}_i$中所有生成的詞元。政策梯度目標計算如下：

$$\mathcal{J}_{\mathrm{RL}}(\theta) = \mathbb{E}_{q\sim\mathcal{D},\{\mathcal{T}_i\}_{i=1}^{N}\sim\pi_{\theta'}(\cdot|q)} \left[\frac{1}{\sum_{i=1}^{N}|\mathcal{T}_i|} \sum_{i=1}^{N}\sum_{t=1}^{|\mathcal{T}_i|} \left(\mathcal{L}_{i,t}^{\mathrm{clip}}(\theta) - \beta D_{\mathrm{KL}}(\pi_\theta||\pi_{\mathrm{ref}}) \right) \right], \quad (5)$$

$$\mathcal{L}_{i,t}^{\mathrm{clip}}(\theta) = \min \left[\gamma_t(\theta)\hat{A}_i, \mathrm{clip}\left(\gamma_t(\theta), 1-\epsilon_{\mathrm{low}}, 1+\epsilon_{\mathrm{high}}\right)\hat{A}_i \right]. \quad (6)$$

其中 $|\mathcal{T}_i|$ 表示軌跡 $\mathcal{T}_i$ 中生成的總 token 數量， $\gamma_t(\theta)$ 代表 π_θ 與採樣策略 $\pi_{\theta'}$ 在步驟 t $A_i = r_i - \mathrm{mean}(R)$ 的概率比值， ϵ_{low} 是群體相對優勢，而 ϵ_{high} 、 [61]則是遵循 DAPO [61]的剪裁超參數。

2.3 VST 的數據合成流程

我們生成了包含影片串流思維數據的集合，用於支援 VST 訓練，這是出於以下原因：現有的思維鏈 (CoT) 數據集大多針對離線視頻大語言模型 (VideoLLMs)，提供整個影片的全局、事後視角，導致在因果串流約束下難以避免資訊洩露。為此，我們引入了一個基於知識圖的自動化數據生成流程。如圖 4 所示，該流程透過串流影片實體提取、證據鏈採樣和串流思維問答合成，生成了具有明確推理路徑的高質量訓練範例。

串流影片實體萃取。為了建立時間上的一致知識圖，我們維護一個實體庫，並從影片串流上的滑動視窗中萃取三元組。我們使用 PySceneDetect 將影片分割成 N 個場景片段。對於每一個輸入的片段，一個離線視頻大語言模型（例如 Gemini 3.0 flash）透過新增新觀察到的實體和關係來更新實體庫

as (**head**, relation, **tail**). When the window exceeds size W, we drop the oldest clip and retain the most recent $W-1$ overlapping clips to preserve temporal continuity. The entity bank thus serves as a lightweight memory for consistent entity tracking and timeline-aligned graph construction.

Evidence Chain Sampling. After processing the whole video, the complete entity bank is refined using an LLM to filter out noise entities, such as duplicates and subtitles. Subsequently, NetworkX [16] is used to construct the knowledge graph, which represents the logical relationships between events in the video. To mine long-term causal dependencies, an initial node is randomly selected, and a depth-first search (DFS) is used to extract evidence chains. Each node in these chains contains detailed information about the head and tail entities, their relationship, timestamps, and scene descriptions, facilitating comprehensive reasoning over the video content. For each video, we sample multiple evidence chains, enforcing that the entity overlap between any two chains is below 10% to promote diversity.

Stream Thought QA Synthesis. The final phase leverages Gemini 3.0 flash as a data synthesizer. Conditioned on the video knowledge graph, the model first generates a streaming CoT rationale to actively reason over video events and dynamic content. Subsequently, aligned with a sampled evidence chain $\{\mathbf{z}^k\}_{k=1}^K$, it synthesizes a query $\mathbf{q}$ and the final answer $\mathbf{y}$, necessitating multi-evidence reasoning that integrates the CoT with visual context. To ensure data fidelity, we apply a strict post-generation filtering rubric, including: world-knowledge check, format alignment, logical consistency, repetition check, and thought validation.

Curation of VST training set. Following the above procedure, we generate 100K streaming-thought examples with videos from LLaVA-Vid [67] and Video-Marathon [31]. In addition, our full supervised fine-tuning corpus for VST-SFT includes 50K open-ended QA instances randomly sampled from LLaVA-Vid. For VST-RL, we train on 11K sampled questions, including multiple-choice questions from LLaVA-Vid, Video-Marathon, and Onethinker [11], as well as counting questions from RepCount [20].

3 Experiment

3.1 Implementation Details

We adopt Qwen2.5-VL [1] as our base offline VideoLLM, processing input videos at 2 fps. Both VST-SFT and VST-RL (7B model) training stages are conducted on 32 × 80GB VRAM GPUs, utilizing the datasets detailed in Sec. 2.3. The visual encoder and projection layer are frozen throughout the entire training process. For VST-SFT, each training sample follows a 128 second time limit,

作為 (主體、關係、尾實體)。當視窗大小超過 W 時，我們會捨棄最老的片段，並保留最新的 $W-1$ 個疊加片段以維持時間上的連續性。因此，實體庫作為一個輕量級的記憶體，用於一致實體追蹤和時間線對齊的圖構建。

證據鏈採樣。在處理完整個影片後，使用大語言模型將完整的實體庫進行精煉，以過濾雜訊實體，例如重複實體和字幕。接著，使用 NetworkX [16] 來構建知識圖，該圖表現影片中事件之間的邏輯關係。為了挖掘長期的因果依賴關係，隨機選擇一個初始節點，並使用深度優先搜索（DFS）來提取證據鏈。這些鏈中的每個節點都包含關於頭實體和尾實體的詳細信息、它們的關係、時間戳和場景描述，促進對影片內容的全面推理。對於每個影片，我們採樣多個證據鏈，強制要求任意兩個鏈之間的實體重疊率低於 10%，以促進多樣性。

串流思考問答合成。最終階段利用 Gemini 3.0 flash 作為數據合成器。基於影片知識圖，模型首先生成一個串流因果推理的推理過程，以主動對影片事件和動態內容進行推理。接著，與採樣的證據鏈 $\{\mathrm{z}^k\}_{k=1}^K$ 對齊，它合成查詢 q 和最終答案 y，需要整合因果推理和視覺內容的多證據推理。為了確保數據忠實性，我們應用嚴格的後生成過濾標準，包括：世界知識檢查、格式對齊、邏輯一致性、重複檢查和思考驗證。

VST 訓練集的策展。遵循上述程序，我們從 LLava-Vid [67] 和 Video-Marathon [31] 中生成 100K 個串流思維範例，搭配影片。此外，我們的 VST-SFT 完整監督微調語料庫包含從 LLava-Vid 隨機抽樣的 50K 個開放式問答範例。對於 VST-RL，我們在 11K 個抽樣問題上進行訓練，包括來自 LLava-Vid、Video-Marathon 和 Onethinker [11], 的選擇題，以及來自 RepCount [20] 的計數題。

3 實驗

3.1 實施細節

我們採用 Qwen2.5-VL [1] 作為我們的基礎離線 VideoLLM，以 2 fps 的速率處理輸入影片。VST-SFT 和 VST-RL (7B 模型) 的訓練階段都在 32 × 80GB VRAM GPU 上進行，利用 Sec. 2.3 中詳細說明的數據集。視覺編碼器和投影層在整個訓練過程中都是凍結的。對於 VST-SFT，每個訓練樣本遵循 128 秒的時間限制，

and overlong raw videos are segmented into clips following Eq. (4). For VST-RL, we employ verl [40] with vLLM [25] and FSDP [68] backend. We configure the rollout batch size to 256 with a group size of $N = 8$, and define the reward function based on the correctness of the final answer. Additionally, following LongVILA-R1 [4], we leverage the paralleled encoding strategy during rollout to pre-compute video embeddings. During testing, following StreamingForest [62], we cap each inference step (including streaming-think and the final answer) at 8,192 video tokens and limit the max thinking times to 4 for efficient evaluation. We conduct all evaluations using the lmms-eval framework [65].

Table 1: Comparison of offline and online VideoLLMs on StreamingBench Real-Time understanding tasks.

Model	Venue	OP	CR	CS	ATP	EU	TR	PR	SU	ACP	CT	**Overall**
Proprietary Models												
Gemini 1.5 pro [43]	-	79.0	80.5	83.5	79.7	80.0	84.7	77.8	64.2	72.0	48.7	75.7
GPT-4o [36]	-	77.1	80.5	83.9	76.5	70.2	83.8	66.7	62.2	69.1	49.2	73.3
Open-source Offline Models												
VILA-1.5-8B [30]	CVPR'24	53.7	49.2	71.0	56.9	53.4	53.9	54.6	48.8	50.1	17.6	52.3
LongVA-7B [66]	TMLR'25	70.0	63.3	61.2	70.9	62.7	59.5	61.1	53.7	54.7	34.7	60.0
MiniCPM-v2.6-7B [21]	COLM'24	71.9	71.1	77.9	75.8	64.6	65.7	70.4	56.1	62.3	53.4	67.4
LLaVA-OV-7B [26]	TMLR'25	80.4	74.2	76.0	80.7	72.7	71.7	67.6	65.5	65.7	45.1	71.1
Qwen2.5-VL-7B [1]	-	78.3	80.5	78.9	80.5	76.7	78.5	79.6	63.4	66.2	53.2	73.7
Open-source Online Models												
Flash-VStream-7B [64]	ICCV'25	25.9	43.6	24.9	23.9	27.3	13.1	18.5	25.2	23.9	48.7	23.2
VideoLLM-online-8B [2]	CVPR'24	39.1	40.1	34.5	31.1	46.0	32.4	31.5	34.2	42.5	27.9	36.0
Dispider-8B [37]	CVPR'25	74.9	75.5	74.1	73.1	74.4	59.9	76.1	62.9	62.2	45.8	67.6
TimeChatOnline-7B [58]	MM'25	80.2	82.0	79.5	83.3	76.1	78.5	78.7	64.6	69.6	58.0	75.4
Streamforest-7B [62]	NeurIPS'25	83.1	82.8	82.7	84.3	77.5	78.2	76.9	69.1	75.6	54.4	77.3
VST-7B (ours)	ECCV'26	85.4	82.0	86.4	89.1	74.2	87.2	82.4	73.1	73.9	47.3	79.5

3.2 Benchmarks

To demonstrate the effectiveness of our method, we conducted a comprehensive evaluation across five video understanding benchmarks. Specifically, StreamingBench [32] and OVO-Bench [35] are utilized for online video understanding, focusing on the model's online reasoning capabilities and temporal awareness. VideoMME [12] serves as a comprehensive offline benchmark covering diverse domains and varying video durations. LongVideoBench [51] is designed to evaluate the long-form video understanding capabilities, while Video-Holmes [5] emphasizes logical reasoning within video content.

3.3 Online Video Benchmark Results

As shown in Tabs. 1 and 2, we evaluate our model on StreamingBench and OVO-Bench. VST-7B achieves 79.5% on StreamingBench and 59.3% on OVO-

超過長度的原始影片會根據公式 (4) 分割成片段。對於 VST-RL，我們採用 verl [40] 與 vLLM [25] 和 FSDP [68] 後端。我們將展開批次大小配置為 256，組大小為 $N = 8$，並根據最終答案的正確性定義獎勵函數。此外，遵循 LongVILA-R1 [4]，我們在展開期間利用並行編碼策略來預計算影片嵌入。在測試期間，遵循 StreamingForest [62]，我們將每個推理步驟（包括串流思考和最終答案）限制在 8,192 個影片 token，並將最大思考次數限制在 4，以進行高效的評估。我們使用 lmms-eval 框架 [65]進行所有評估。

表1：比較在StreamingBench實時理解任務上線上與線下VideoLLMs的表現。

模型	場地	OP	CR	CS	ATP	EU	TR	PR	SU	ACP	CT	**整體**
專有模型												
Gemini 1.5 pro [43]	-	79.0	80.5	83.5	79.7	80.0	84.7	77.8	64.2	72.0	48.7	75.7
GPT-4o [36]	-	77.1	80.5	83.9	76.5	70.2	83.8	66.7	62.2	69.1	49.2	73.3
開源離線模型												
VILA-1.5-8B [30]	CVPR' 24	53.7	49.2	71.0	56.9	53.4	53.9	54.6	48.8	50.1	17.6	52.3
LongVA-7B [66]	TMLR' 25	70.0	63.3	61.2	70.9	62.7	59.5	61.1	53.7	54.7	34.7	60.0
MiniCPM-v2.6-7B [21]	COLM' 24	71.9	71.1	77.9	75.8	64.6	65.7	70.4	56.1	62.3	53.4	67.4
LLaVA-OV-7B [26]	TMLR' 25	80.4	74.2	76.0	80.7	72.7	71.7	67.6	65.5	65.7	45.1	71.1
Qwen2.5-VL-7B [1]	-	78.3	80.5	78.9	80.5	76.7	78.5	79.6	63.4	66.2	53.2	73.7
開源線上模型												
Flash-VStream-7B [64]	ICCV' 25	25.9	43.6	24.9	23.9	27.3	13.1	18.5	25.2	23.9	48.7	23.2
VideoLLM-online-8B [2]	CVPR' 24	39.1	40.1	34.5	31.1	46.0	32.4	31.5	34.2	42.5	27.9	36.0
Dispider-8B [37]	CVPR' 25	74.9	75.5	74.1	73.1	74.4	59.9	76.1	62.9	62.2	45.8	67.6
TimeChatOnline-7B [58]	MM' 25	80.2	82.0	79.5	83.3	76.1	78.5	78.7	64.6	69.6	58.0	75.4
Streamforest-7B [62]	NeurIPS' 25	83.1	82.8	82.7	84.3	77.5	78.2	76.9	69.1	75.6	54.4	77.3
VST-7B (我們的)	ECCV' 26	85.4										79.5

3.2 基準測試

為了展現我們方法的效能，我們在五個影片理解基準測試上進行了全面評估。具體來說，StreamingBench [32] 和 OVO-Bench [35] 用於線上影片理解，重點在於模型的線上推理能力與時間感知。VideoMME [12] 作為一個涵蓋多樣領域與不同影片時長的綜合性離線基準測試。LongVideoBench [51] 設計用於評估長片段影片理解的能力，而 Video-Holmes [5] 則強調影片內容中的邏輯推理。

3.3 線上影片基準測試結果

如 Tab. 1 和 2 所示，我們在 StreamingBench 和 OVO-Bench 上評估我們的模型。VST-7B 在 StreamingBench 上達到 79.5%，在 OVO-

Bench, clearly outperforming prior open-source streaming SOTA models, including Streamforest [62] (77.3%) on StreamingBench and Streamo [53] (57.9%) on OVO-Bench. Notably, despite being much smaller than proprietary models, our method surpasses GPT-4o and Gemini 1.5 pro on StreamingBench by 6.2% and 3.8%, respectively, and achieves comparable performance with GPT-4o on OVO-Bench. Beyond the overall scores, VST-7B is particularly strong on OVO-Bench's Backward Tracing task, where it achieves 56.7%, outperforming Streamforest by +4.7%. This result indicates that our model can retain and retrieve historical information effectively, supporting sustained memory over streaming inputs. These results highlight the strength of our approach for streaming video understanding. We believe the gains stem from our VST paradigm and a tailored post-training recipe, which together improve the model's ability.

Table 2: Comparison of offline and online VideoLLMs on OVO-Bench.

Model	Venue	Real-Time							Backward				Forward				**Overall**
		OCR	ACR	ATR	STU	FPD	OJR	Avg.	EPM	ASI	HLD	Avg.	REC	SSR	CRR	Avg.	Avg.
Proprietary Models																	
Gemini 1.5 pro [43]	-	85.9	67.0	79.3	58.4	63.4	62.0	69.3	58.6	76.4	52.6	62.5	35.5	74.2	61.7	57.2	63.0
GPT-4o [36]	-	69.8	64.2	71.6	51.1	70.3	59.8	64.5	57.9	75.7	48.7	60.8	27.6	73.2	59.4	53.4	59.5
Open-source Offline Models																	
Qwen2-VL-72B [47]	-	65.8	60.6	69.8	51.7	69.3	54.4	61.9	52.5	60.8	57.5	57.0	38.8	64.1	45.0	49.3	56.3
LLaVA-Video-7B [67]	TMLR'25	69.1	58.7	68.8	49.4	74.3	59.8	63.5	56.2	57.4	7.5	40.4	34.1	70.0	60.4	54.8	52.9
LLaVA-OV-7B [26]	TMLR'25	66.4	57.8	73.3	53.4	71.3	62.0	64.0	54.2	55.4	21.5	43.7	25.6	67.1	58.8	50.5	52.7
LongVU-7B [39]	ICML'25	53.7	53.2	62.9	47.8	68.3	59.8	57.6	40.7	59.5	4.8	35.0	12.2	69.5	60.8	47.5	46.7
Open-source Online Models																	
VideoLLM-online-8B [2]	CVPR'24	8.1	23.9	12.1	14.0	45.5	21.2	20.8	22.2	18.8	12.2	17.7	-	-	-	-	-
Dispider-8B [37]	CVPR'25	57.7	49.5	62.1	44.9	61.4	51.6	54.6	48.5	55.4	4.3	36.1	18.1	37.4	48.8	34.7	41.8
TimeChatOnline-7B [58]	MM'25	75.2	46.8	70.7	47.8	69.3	61.4	61.9	55.9	59.5	9.7	41.7	31.6	38.5	40.0	36.7	46.7
Streamforest-7B [62]	NeurIPS'25	68.5	53.2	71.6	47.8	65.4	60.9	61.2	58.9	64.9	32.3	52.0	32.8	70.6	57.1	52.5	55.6
Streamo-7B [53]	CVPR'26	77.2	66.1	76.7	45.5	66.3	72.8	67.4	55.6	58.1	33.9	49.2	30.8	57.6	82.5	57.0	57.9
VST-7B (ours)	ECCV'26	80.5	55.1	72.4	55.1	76.2	64.1	67.2	56.9	64.9	48.4	56.7	33.0	66.9	62.1	54.0	59.3

3.4 Offline Video Benchmark Results

In Tab. 3, we evaluate VST-7B on three offline video benchmarks, including VideoMME, LongVideoBench, and VideoHolmes. The results show that VST-7B delivers competitive performance across all three datasets, with particularly strong gains on long-video understanding and complex reasoning. On long-video benchmarks, VST-7B achieves 55.3% on VideoMME-long, outperforming TimeChat-Online by +6.9%, and 58.0% on LongVideoBench, exceeding it by +2.6%. On the reasoning benchmark VideoHolmes, VST-7B reaches 41.9%, surpassing Video-R1 by +5.4%. We attribute these improvements to our streaming thinking framework, which enables dynamic thinking over long videos to build long-term memory, and leverages both historical memory and current visual context for deep reasoning.

Bench 上達到 59.3%，明顯優於先前的開源串流 SOTA 模型，包括 Streamforest [62] (77.3%) 在 StreamingBench 上和 Streamo [53] (57.9%) 在 OVO-Bench 上。值得注意的是，儘管比我們的專有模型小得多，我們的方法在 StreamingBench 上分別超過 GPT-4o 和 Gemini 1.5 pro 6.2% 和 3.8%，並在 OVO-Bench 上達到與 GPT-4o 相當的性能。除了整體分數外，VST-7Bis 在 OVO-Bench 的反向追蹤任務上特別強勁，達到 56.7%，超過 Streamforest +4。7%。這個結果表明我們的模型可以有效地保留和檢索歷史信息，支持在串流輸入上的持續記憶。這些結果突顯了我們方法在串流視頻理解方面的優勢。我們相信這些進步源於我們的 VST 范式和定制的訓練後配方，它們共同提高了模型的能力。

表 2：OVO-Bench 上離線與線上 VideoLLMs 的比較。

模型	場地	實時							反向				正向				**整體**
		OCR	ACR	ATR	STU	FPD	OJR	Avg.	EPM	ASI	HLD	Avg.	REC	SSR	CRR	Avg.	Avg.
專有模型																	
Gemini 1.5 pro [43]	-	85.9	67.0	79.3	58.4	63.4	62.0	69.3	58.6	76.4	52.6	62.5	35.5	74.2	61.7	57.2	63.0
GPT-4o [36]	-	69.8	64.2	71.6	51.1	70.3	59.8	64.5	57.9	75.7	48.7	60.8	27.6	73.2	59.4	53.4	59.5
開源離線模型																	
Qwen2-VL-72B [47]	-	65.8	60.6	69.8	51.7	69.3	54.4	61.9	52.5	60.8	57.5	57.0	38.8	64.1	45.0	49.3	56.3
LLaVA-Video-7B [67]	TMLR' 25	69.1	58.7	68.8	49.4	74.3	59.8	63.5	56.2	57.4	7.5	40.4	34.1	70.0	60.4	54.8	52.9
LLaVA-OV-7B [26]	TMLR' 25	66.4	57.8	73.3	53.4	71.3	62.0	64.0	54.2	55.4	21.5	43.7	25.6	67.1	58.8	50.5	52.7
LongVU-7B [39]	ICML' 25	53.7	53.2	62.9	47.8	68.3	59.8	57.6	40.7	59.5	4.8	35.0	12.2	69.5	60.8	47.5	46.7
開源線上模型																	
VideoLLM-online-8B [2]	CVPR' 24	8.1	23.9	12.1	14.0	45.5	21.2	20.8	22.2	18.8	12.2	17.7	-	-	-	-	-
Dispider-8B [37]	CVPR' 25	57.7	49.5	62.1	44.9	61.4	51.6	54.6	48.5	55.4	4.3	36.1	18.1	37.4	48.8	34.7	41.8
TimeChatOnline-7B [58]	MM' 25	75.2	46.8	70.7	47.8	69.3	61.4	61.9	55.9	59.5	9.7	41.7	31.6	38.5	40.0	36.7	46.7
Streamforest-7B [62]	NeurIPS' 25	68.5	53.2	71.6	47.8	65.4	60.9	61.2	58.9	64.9	32.3	52.0	32.8	70.6	57.1	52.5	55.6
Streamo-7B [53]	CVPR' 26	77.2	66.1	76.7	45.5	66.3	72.8	67.4	55.6	58.1	33.9	49.2	30.8	57.6	82.5	57.0	57.9
VST-7B (我們的)	ECCV' 26	80.5															59.3

3.4 線上影片基準測試結果

在表3中，我們在VideoMME、LongVideoBench和VideoHolmes三個離線影片基準測試上評估VST-7B。結果顯示，VST-7B在所有三個數據集上都展現出競爭力，特別是在長影片理解和複雜推理方面有顯著提升。在長影片基準測試中，VST-7B在VideoMME-long上達到55.3%，超過TimeChat-Online +6.9%，在LongVideoBench上達到58.0%，超過它 +2.6%。在推理基準測試VideoHolmes上，VST-7B達到41.9%，超過Video-R1 +5.4%。我們將這些改進歸因於我們的串流思維框架，該框架能夠在長影片上進行動態思考以建立長期記憶，並利用歷史記憶和當前視覺語境進行深度推理。

Table 3: Comparison of offline and online VideoLLMs on VideoMME (without subtitles), LongVideoBench, and VideoHolmes.

Model	Venue	VideoMME w/o sub.		LongVideoBench	VideoHolmes
		Long	Overall		
Proprietary Models					
Gemini 1.5 pro [43]	-	67.4	75.0	64.0	45.7
GPT-4o [36]	-	65.3	71.9	66.7	42.0
Open-source Offline Models					
LongVA-7B [66]	TMLR'25	47.6	54.3	56.3	-
Video-R1-7B [10]	NeurIPS'25	-	61.4	-	36.5
LongVILA-R1-7B [4]	NeurIPS'25	55.2	65.1	58.0	-
REVISOR-7B [27]	CVPR'26	56.2	65.7	57.5	-
Open-source Online Models					
Dispider-7B [37]	CVPR'25	-	57.2	-	-
Streamforest-7B [62]	NeurIPS'25	-	61.4	-	-
TimeChatOnline-7B [58]	MM'25	48.4	62.4	55.4	-
VST-7B (Ours)	ECCV'26	55.3	64.9	58.0	41.9

Table 4: Ablation study on VST training schedule.

Model & Config	OVO-Bench			VideoMME
	Backward	Forward	Overall	w/o sub. Overall
Qwen2.5-VL-7B (Base model)	47.5	41.9	50.5	62.9
Ablation on VST-SFT training data				
+LLava-Vid (50K)	49.9	42.4	52.3	61.8
+LLava-Vid (30K) & VST (20K)	52.0	50.1	56.8	62.5
+LLava-Vid (20K) & VST (30K)	53.3	50.0	57.1	63.1
Ablation on different training stage				
+VST-SFT	56.7	48.5	57.4	63.0
+VST-RL	49.3	54.6	56.8	62.8
+VST-SFT & VST-RL	56.7	54.0	59.3	64.9

3.5 Ablation Study

Ablation on training schedule. As shown in Tab. 4, we first analyze the composition of the SFT training data. Mixing our VST data with the LLaVA-Vid QA dataset consistently improves online video understanding. Specifically, compared to using 50K LLaVA-Vid data alone, the mix of 20K LLaVA-Vid and 30K VST data achieves a +4.8% gain on the OVO-Bench. Furthermore, the ablation on different training stages demonstrates that our training strategies effectively enhance online video capabilities. Interestingly, we find that VST-SFT primarily benefits the model's backward memory capacity (+9.2% improvement on Backward track), while VST-RL is advantageous for forward prediction capabilities (improving the Forward score of +12.7%). Finally, combining both stages (VST-SFT & VST-RL) yields the highest overall performance on both OVO-Bench (59.3%) and VideoMME (64.9%).

表3：比較離線與線上VideoLLMs在VideoMME（無字幕）、LongVideoBench和VideoHolmes上的表現。

模型	場地	VideoMME 無字幕		LongVideoBench	VideoHolmes
		Long	整體		
專有模型					
Gemini 1.5 pro [43]	-	67.4	75.0	64.0	45.7
GPT-4o [36]	-	65.3	71.9	66.7	42.0
開源離線模型					
LongVA-7B [66]	TMLR' 25	47.6	54.3	56.3	-
Video-R1-7B [10]	NeurIPS' 25	-	61.4	-	36.5
LongVILA-R1-7B [4]	NeurIPS' 25	55.2	65.1	58.0	-
REVISOR-7B [27]	CVPR' 26	56.2	65.7	57.5	-
開源線上模型					
Dispider-7B [37]	CVPR' 25	-	57.2	-	-
Streamforest-7B [62]	NeurIPS' 25	-	61.4	-	-
TimeChatOnline-7B [58]	MM' 25	48.4	62.4	55.4	-
VST-7B (我們的)	ECCV' 26	55.3	64.9	58.0	41.9

表4：關於VST訓練時間表的消融研究。

模型 & 設定	OVO-Bench			VideoMME
	反向	正向	整體	無字幕 整體
Qwen2.5-VL-7B (基礎模型)	47.5	41.9	50.5	62.9
對 *VST-SFT* 訓練數據進行刪除				
+LLava-Vid (50K)	49.9	42.4	52.3	61.8
+LLava-Vid (30K) & VST (20K)	52.0	50.1	56.8	62.5
+LLava-Vid (20K) & VST (30K)	53.3	50.0	57.1	63.1
不同訓練階段的消融研究				
+VST-SFT	56.7	48.5	57.4	63.0
+VST-RL	49.3	54.6	56.8	62.8
+VST-SFT & VST-RL	56.7	54.0	59.3	64.9

3.5 消融研究

訓練計劃的消融實驗。如表4所示，我們首先分析SFT訓練數據的組成。將我們的VST數據與LLaVA-Vid問答數據集混合能夠一致性地提升線上影片理解能力。具體來說，與僅使用50K LLaVA-Vid數據相比，混合20K LLaVA-Vid和30K VST數據在OVO-Bench上實現了 +4.8%的提升。此外，不同訓練階段的消融實驗表明，我們的訓練策略有效地增強了線上影片能力。有趣的是，我們發現VST-SFT主要受益於模型的向後記憶能力 (+9.2%的提升在Backward track上)，而VST-RL則有助於向前預測能力（提升了向前得分的 +12.7%）。最終，結合兩個階段（VST-SFT & VST-RL）在OVO-Bench（59.3%）和VideoMME（64.9%）上都取得了最高的整體性能。

Table 5: Ablation study on the scale of the base offline VideoLLM. All results in this table are obtained with the VST inference pipeline, where Qwen is evaluated as a zero-shot inference baseline.

Size	Model	OVOB. Overall	StreamingB. Realtime	V-MME w/o sub. Overall	LongVideoB.	VideoHolmes
3B	Qwen2.5-VL	53.1	67.8	57.9	53.3	30.7
	VST	56.2	75.5	59.5	54.1	36.1
7B	Qwen2.5-VL	55.0	71.7	62.3	54.7	32.9
	VST	59.3	79.5	64.9	58.0	41.9
32B	Qwen2.5-VL	60.1	71.5	65.8	59.8	40.1
	VST	63.5	80.7	67.2	60.7	45.1

Ablation on Streaming Thinking Times at Inference. Fig. 5 analyzes the impact of maximum streaming thinking times on OVO-Bench. For the Backward task, accuracy increases from 53.3% and grows continuously from 1 to 16 steps, ultimately reaching 57.5%. This demonstrates that additional thinking steps help generate precise memories for backward tracing. For the Real-Time and Forward tasks, initial thinking steps clearly aid in understanding visual information. However, performance reaches a plateau for ≥ 4 steps, as excessive memory details introduce redundancy.

Ablation on Base Model Size. Tab. 5 examines the impact of the base model capacity. Note that the base Qwen rows here are evaluated under the VST streaming inference pipeline (zero-shot), and thus differ from the native offline base in Tab. 4; this isolates the contribution of training from that of the inference protocol. We apply our two-stage training recipe (VST-SFT and VST-RL) to the Qwen2.5-VL-Instruct models at 3B, 7B, and 32B scales. The Video Streaming Thinking paradigm yields consistent improvements across all online and offline benchmarks regardless of the model size. For instance, on StreamingBench, VST achieves absolute accuracy gains of +7.7%, +7.8%, and +9.2% over the 3B, 7B, and 32B base models, respectively. Similar consistent enhancements are observed on complex tasks like VideoHolmes (+5.4%, +9.0%, and +5.0%). These results demonstrate that our proposed method is highly parameter-scalable.

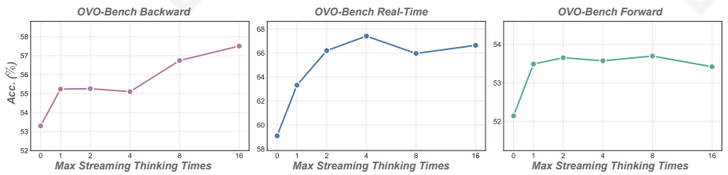


Fig. 5: Ablation study on max thinking times.

表5：基礎離線VideoLLM規模的消融研究。本表所有結果均使用VST推論管線獲得，其中Qwen被評估為零樣本推論基線。

Size	模型	OVOB. 整體	串流B. 實時	V-MME 無字幕 整體	LongVideoB.	VideoHolmes
3B	Qwen2.5-VL	53.1	67.8	57.9	53.3	30.7
	VST	56.2	75.5	59.5	54.1	36.1
7B	Qwen2.5-VL	55.0	71.7	62.3	54.7	32.9
	VST	59.3	79.5	64.9	58.0	41.9
32B	Qwen2.5-VL	60.1	71.5	65.8	59.8	40.1
	VST	63.5	80.7	67.2	60.7	45.1

在推論中對串流思考時間進行消融研究。圖5分析了最大串流思考時間對OVO-Bench的影響。對於反向任務，準確率從53.3%開始，持續從1到16步增長，最終達到57.5%。這表明額外的思考步驟有助於為反向追蹤生成精確的記憶。對於實時和正向任務，初始思考步驟顯然有助於理解視覺信息。然而，性能在 ≥ 4 步驟達到平台期，因為過多的記憶細節會引入冗餘。

基礎模型大小上的刪除實驗。表5檢驗了基礎模型容量的影響。請注意，這裡的基礎Qwen行是基於VST串流推理流程（零樣本）進行評估的，因此與表4中的原生離線基礎模型不同；這可以隔離訓練貢獻與推理協議貢獻。我們將我們的兩階段訓練配方（VST-SFT和VST-RL）應用於3B、7B和32B規模的Qwen2.5-VL-Instruct模型。視頻串流思考范式在所有線上和離線基準上均產生了一致性的改進，無論模型大小如何。例如，在StreamingBench上，VST在3B、7B和32B基礎模型上分別實現了 +7.7%、 +7.8%和 +9.2%的絕對準確率提升。在複雜任務如Video Holmes上，也觀察到類似的持續增強，分別為+5.4%、 +9.0%和 +5.0%。這些結果證明我們提出的方法具有高度參數可扩展性。

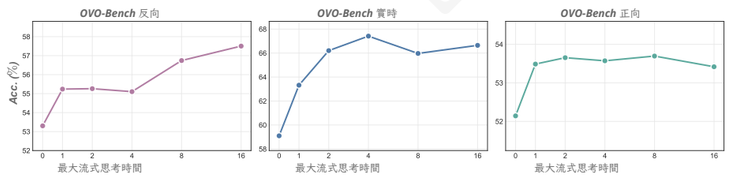


圖5：最大思考時間的消融研究。

3.6 Analysis

Efficiency Analysis. We compare the QA latency of several offline and online methods under the same experimental setup. All measurements are conducted on VideoHolmes, as shown in Tab. 6. Models without CoT directly output the final answer. Benefiting from our query-ahead streaming think mechanism, VST maintains significantly lower response latency. Moreover, streaming think is executed asynchronously before the query and finishes within the clip inter-arrival interval, so its computation is amortized over playback rather than added after the query. As a result, it does not increase the real-world end-to-end inference time.

Table 6: Inference Latency.

Online / Offline Method	**QA Latency**
Qwen2.5-VL-7B	0.54s
Qwen2.5-VL-7B w/CoT	5.30s
Video-R1 w/CoT	8.80s
VideoLLM-online-8B	0.38s
Dispider-7B	1.10s
VST-3B (Ours)	0.53s
VST-7B (Ours)	0.56s
VST-32B (Ours)	1.40s

In our implementation, streaming thoughts are generated asynchronously every 16–32 seconds and take 7.0 seconds on average, with a P99 latency of 11.2 seconds. Since this is shorter than the minimum trigger interval, the overhead is amortized over playback. If interrupted by a query or a new step, VST uses the latest completed memory state. Therefore, VST adds background computation but not post-query response latency.

Case Study. Fig. 6 presents a case study from VideoHolmes. The query requires temporal reasoning over disjoint segments, specifically aligning repeated visualizations of a wall clock with the subsequent appearance of a "blurred-face man". The baseline Video-R1-7B, which relies on post-query thinking, fails to capture these dispersed temporal cues due to the difficulty of attending to specific evidence across a long context. Consequently, it hallucinates a spurious correlation involving object interactions, leading to a logical error. Furthermore, this retrospective reasoning incurs a significant latency of 9.53s. In contrast, VST-7B employs streaming thinking to continuously update its evidence (e.g., timestamps and event triggers) as the video memories. This pre-query evidence accumulation allows VST to correctly deduce the time-based rule and, by shifting the reasoning burden to the streaming phase, drastically reduces the response latency to 0.51s. This comparison demonstrates that pre-query streaming thinking simultaneously enhances reasoning robustness and system responsiveness.

We further examine typical failure cases of VST during inference. While streaming thinking improves evidence accumulation, the remaining errors are mainly associated with imperfect memory construction. Specifically, VST may store visually salient but query-irrelevant details, compress early evidence too aggressively, miss fine-grained temporal spans, or fail to associate weak cues across distant events. These observations suggest that more fine-grained and faithful memory construction, especially for preserving subtle temporal evidence and cross-event relations, is a promising direction for future work.

3.6 分析

效率分析。我們在相同的實驗設置下比較了多種離線和線上方法的問答延遲。所有測量均在VideoHolmes上進行，如表6所示。沒有CoT的模型直接輸出最終答案。受益於我們的查詢預先串流思考機制，VST維持顯著更低的響應延遲。此外，串流思考在查詢之前異步執行

表 6：推論延遲。

線上 / 離	**問答延遲**
Qwen2.5-VL-7B	0.54s
Qwen2.5-VL-7B w/因果推理	5.30s
Video-R1 w/因果推理	8.80s
VideoLLM線上-8B	0.38s
Dispider-7B	1.10s
VST-3B (我們的)	0.53s
VST-7B (我們的)	0.56s
VST-32B (Ours)	1.40s

完成於片段內的到達間隔內，因此其計算是分攤在播放而非查詢之後。結果，它不增加實際的端到端推理時間。

在我們的實現中，串流思考每 16–32 秒非同步產生，平均耗時 7.0 秒，P99 延遲為 11.2 秒。由於這比最小觸發間隔短，因此開銷是分攤在播放上。如果被查詢或新步驟中斷，VST 使用最新完成的記憶狀態。因此，VST 增加了背景計算，但不是查詢後的響應延遲。

案例研究。圖 6 來自 VideoHolmes 的案例研究。查詢需要跨不同片段的時間推理，具體是對齊重複出現的牆上鐘與隨後「模糊臉孔的人」出現。依賴查詢後思考的基線 Video-R1-7B，由於難以在長文本中專注於特定證據，無法捕捉這些分散的時間線索。結果，它虛構了一個涉及物體互動的假相關性，導致邏輯錯誤。此外，這種回顧性推理產生了 9.53s 的顯著延遲。相比之下，VST-7B 則採用串流思考來持續更新其證據（例如，時間戳和事件觸發器）作為影片記憶。這種查詢前證據積累使 VST 能正確推斷時間規則，並將推理負擔轉移到串流階段，將響應延遲顯著減少至 0.51s。這個比較說明查詢前串流思考同時增強了推理穩健性和系統響應性。

我們進一步檢驗了 VST 在推理過程中典型的失敗案例。雖然串流思考能夠提升證據積累，但剩餘的錯誤主要與不完美的記憶構建有關。具體來說，VST 可能會儲存視覺上顯著但查詢無關的細節，過於激進地壓縮早期證據，錯過細粒度的時間跨度，或無法將遠距離事件之間的微弱線索聯繫起來。這些觀察表明，更細粒度且忠於事實的記憶構建——特別是為了保留微妙的時間證據和跨事件關係——是未來工作的有前景的方向。

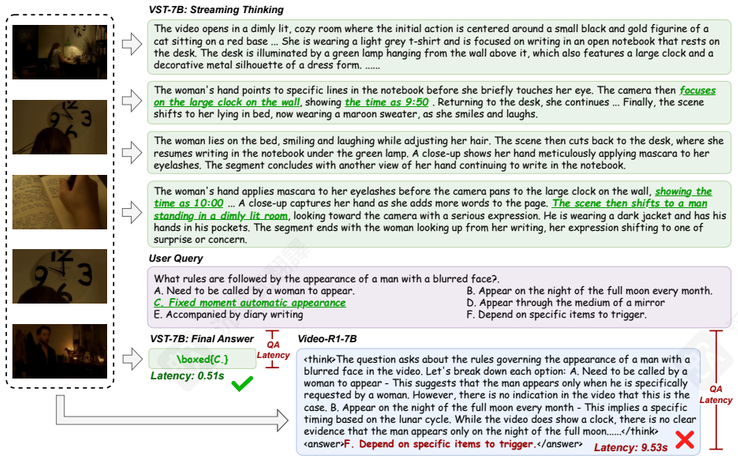


Fig. 6: Case Study from VideoHolmes. We compare VST-7B with Video-R1-7B. VST-7B processes the video stream and performs streaming thinking before the query, then answers directly once the query arrives. In contrast, Video-R1-7B generates CoT after the query, resulting in higher QA latency. VST-7B achieves better performance with lower QA latency in this example.

4 Related Work

Streaming Video Understanding. Streaming video understanding processes continuous visual inputs of indeterminate length. Unlike offline methods, the lack of global sampling and restricted context windows poses significant challenges for VideoLLMs. Some existing methods attempt to retain extended video information within limited context lengths through real-time visual token compression [2,38,41,55,58,62]. For instance, rLiVS reduces streaming token redundancy by recurrently pruning attention-uninformative visual tokens, echoing broader attention-reuse strategies for efficient visual processing [7, 59]. Others incorporate external memory mechanisms to recall historical information via similarity ranking [6,37,57,64]. However, these methods rely on static heuristics, lacking autonomous memory management and the ability to perform complex, multi-step reasoning. To bridge this gap, recent studies have introduced streaming and long-context reasoning methods into video understanding [46,50]. Along this line, this paper proposes ***V**ideo **S**treaming **T**hinking* (VST), which introduces an online thinking process that evolves with the video stream. By coupling autonomous memory management with in-depth instruction analysis, VST enables models to transcend short-range perception and achieve robust streaming intelligence.



圖6：來自VideoHolmes的案例研究。我們比較VST-7B與Video-R1-7B。VST-7B處理影片串流，在查詢前進行串流思考，查詢到達後直接回答。相比之下，Video-R1-7B在查詢後才生成因果推理，導致問答延遲更高。在本例中，VST-7B以較低的問答延遲達成更好的性能。

4 相關研究

串流視頻理解。串流視頻理解處理無法確定長度的連續視覺輸入。與離線方法不同，缺乏全局採樣和受限的上下文窗口對視頻大語言模型（VideoLLMs）構成顯著挑戰。一些現有方法嘗試通過實時視覺標記壓縮，在有限的上下文長度內保留擴展的視頻信息 [2,38,41,55,58,62]。例如，rLiVS透過反覆刪減無信息的視覺標記來減少串流標記冗餘，反映了為高效視覺處理而設計的更廣泛的注意力重用策略 [7, 59]。其他人則整合外部記憶機制，透過相似性排名來回憶歷史信息 [6,37,57,64]。然而，這些方法依賴靜態啟发式方法，缺乏自主記憶管理和執行複雜、多步推理的能力。為了彌合這一差距，最近的研究將串流和長上下文推理方法引入視頻理解 [46,50]。沿著這條路線，本文提出了視頻串流思考（VST），它引入了一個隨視頻串流演變的線上思考過程。透過結合自主記憶管理與深入指令分析，VST使模型能夠超越短範圍感知，並實現穩固的串流智能。

VideoLLMs Test-Time Scaling. Following the breakthrough of test-time scaling and chain-of-thought in LLMs [15,44,49], recent VideoLLMs have adopted supervised fine-tuning (SFT) to mimic expert reasoning trajectories [17, 18] or utilized R1-style reinforcement learning (RL) to enhance task performance [4, 10, 23, 27, 48, 56]. Despite these advances, existing post-training research remains predominantly confined to offline video understanding. The exploration of reasoning within streaming contexts, particularly regarding long-horizon cognitive capabilities, remains a critical yet neglected frontier. In this paper, we introduce a unified SFT and RL framework for streaming video understanding. Our method achieves a synergistic balance between real-time responsiveness and sophisticated reasoning, enabling autonomous memory management and in-depth analysis of evolving video streams.

5 Conclusion

In this paper, we propose ***V**ideo **S**treaming **T**hinking* (VST), a new paradigm for streaming video understanding that introduces a synchronized stream of logical inference with real-time responsiveness. VST enables a *thinking-while-watching* mechanism that performs reasoning over incoming clips during streaming. We further develop a post-training recipe (VST-SFT and VST-RL) and an automated data synthesis pipeline based on video knowledge graphs to produce streaming-thought supervision. Empirically, VST delivers strong performance across multiple online and offline video understanding benchmarks, and yields consistent gains across VideoLLMs ranging from 3B to 32B parameters, indicating that the approach generalizes across model scales. Overall, our study establishes VST as a practical test-time scaling approach for streaming scenarios, simultaneously enabling explicit CoT generation and real-time responsiveness.

Limitation and Future Works. While the computation of streaming thoughts can be scheduled in parallel with incoming video clips, the additional LLM token consumption is still non-negligible. A promising direction is to explore *latent reasoning* to enable more token-efficient streaming thinking. Moreover, VST primarily focuses on text-guided memory management, which is orthogonal to existing streaming visual memory mechanisms. Investigating their combination and potential synergy is an interesting avenue for future work.

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視頻大語言模型（VideoLLMs）測試時刻縮放。在測試時刻縮放和思維鏈在大語言模型（LLMs） [15,44,49], 取得突破後，最近的視頻大語言模型（VideoLLMs）採用了監督微調（SFT）來模擬專家推理軌跡 [17, 18] ，或利用R1風格強化學習（RL）來提升任務性能 [4,10, 23, 27, 48, 56]。儘管取得了這些進步，現有的後訓練研究仍然主要限於離線視頻理解。在串流環境中進行推理的探索，特別是關於長時程認知能力，仍然是一個關鍵但被忽視的前沿。在本文中，我們引入了一個統一的監督微調（SFT）和強化學習（RL）框架，用於串流視頻理解。我們的方法在實時響應和複雜推理之間達到了協同平衡，實現了自主記憶管理和對演變視頻串流的深入分析。

5 結論

在這篇論文中，我們提出了視頻串流思考（VST），一種新的串流視頻理解範式，它引入了同步的邏輯推理串流與實時響應能力。VST實現了一種邊看邊思考的機制，在串流過程中對接收到的影片片段進行推理。我們進一步開發了一種訓練後配方（VST-SFT和VST-RL）以及基於視頻知識圖的自動化數據合成流程，以產生串流思考監督。實驗上，VST在多個線上和離線視頻理解基準上展現出強大的性能，並在從3B到32B參數的VideoLLMs上產生持續的增益，表明该方法可以泛化到不同的模型規模。整體而言，我們的研究將VST建立為一種實用的測試時規模擴展方法，同時實現了明確CoT生成與實時響應能力。

限制與未來工作。雖然串流思考的計算可以與接收到的影片片段並行安排，但額外的LLM token消耗仍然不可忽視。一個有前景的方向是探索潛在推理，以實現更token高效的串流思考。此外，VST主要關注文字導向記憶管理，這與現有的串流視覺記憶機制是正交的。調查它們的組合和潛在的協同作用是一個有趣的未來工作方向。

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6 Details of VST Inference

6.1 Inference Prompt

We detail the inference prompts used for ***V**ideo **S**treaming **T**hinking* (VST). Both the training and inference phases strictly follow this format. As discussed in the Method part, the LLM generates two distinct types of responses: 1) intermittent streaming inference as the video progresses, which is conditioned primarily on past memory and the current video clip (Tab. 7, top); and 2) final answer generation upon receiving the user query, which is conditioned on the accumulated memory, the current video clip, and the specific question (Tab. 7, bottom).

Table 7: Template of VST inference for streaming thinking (top part) and final answer generation (bottom). Curly-brace placeholders **{}** will be replaced with actual content. {Memory} denotes the historical outputs from previous streaming thinking, comprising the corresponding timestamps and generated content, while {VideoClip} represents the incoming video stream.

```
[System]
You are a Streaming Video Analyst.
{Memory}
{TimeStamp} {VideoClip}
```

```
[System]
You are a Streaming Video Analyst.
{Memory}
{TimeStamp} {VideoClip}
{QueryTime} Based on the provided Video Memory and the Current Video
Clip, answer the following Problem.
{Problem}
Output the final answer in \boxed{}
Your answer:
```

6.2 Streaming Inference Pipeline

Fig. 7 illustrates the streaming reasoning pipeline of VST. Prior to receiving the user query, we conduct a streaming thinking process for each video clip, ensuring the output is generated before the subsequent clip arrives. Consequently, our method effectively utilizes the natural waiting time inherent in real-world video streams. This enables a rapid response once the user poses a question, where the QA latency is defined as the time elapsed from the user's query submission to the LLM's response.

6 視頻串流思考 (VST) 推理細節

6.1 推理提示

我們詳細說明了用於視頻串流思考 (VST) 的推理提示。訓練和推理階段都嚴格遵循此格式。如方法部分所述，大語言模型 (LLM) 生成兩種不同的響應類型：1) 隨著影片推進的間歇式串流推理，主要依據過去記憶和當前影片片段（表7，上方）；以及2) 在接收使用者查詢後生成的最終答案，依據累積的記憶、當前影片片段和具體問題（表7，下方）。

表7：視頻串流思考 (VST) 推理的模板（上方部分）和最終答案生成（下方）。大括號占位符 {} 將被替換為實際內容。{記憶} 表示來自先前串流思考的歷史輸出，包括相應的時間戳和生成內容，而 {影片片段} 則代表即將輸入的影片串流。

```
[System]
You are a Streaming Video Analyst.
{Memory}
{TimeStamp} {VideoClip}
```

```
[System]
You are a Streaming Video Analyst.
{Memory}
{TimeStamp} {VideoClip}
{QueryTime} Based on the provided Video Memory and the Current Video
Clip, answer the following Problem.
{Problem}
Output the final answer in \boxed{}
Your answer:
```

6.2 串流推理流程

圖7說明了VST的串流推理流程。在收到使用者查詢之前，我們會針對每個影片片段進行串流思考過程，確保輸出在下一個片段到達前就已生成。因此，我們的方法能有效利用真實世界影片串流中固有的自然等待時間。這使得當使用者提出問題時，能夠快速回應，其中問答延遲定義為使用者查詢提交到LLM的回應之間經過的時間。

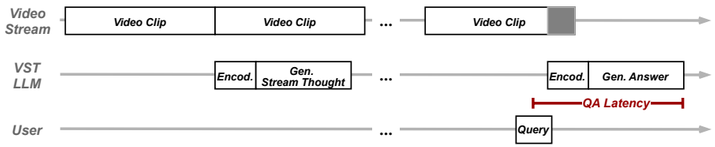


Fig. 7: The streaming inference pipeline of VST. By generating stream thoughts for incoming video clips before a user query arrives, VST effectively hides reasoning latency and enables rapid QA responses.

7 Details of VST-SFT / RL Training

VST-SFT Hyperparameters. To train a streaming model capable of continuous thinking and ensure reproducibility, we report the hyperparameters for the VST-SFT phase. We sample up to 384 frames per video and limit the maximum video pixels to 19,267,584, capping video tokens at 24K to reserve 8K context for language and reasoning. The model is trained for 1 epoch with a learning rate of 5e-6 and 8 gradient accumulation steps.

VST-RL Hyperparameters. For the reinforcement learning phase, we employ the GRPO algorithm to optimize the model. The maximum prompt length is limited to 11,000 tokens, with 1,000 reserved for the maximum generated response length. During the rollout phase, we generate 8 candidate responses per prompt using vLLM with a temperature of 1.0 and a top-p of 0.98. The model is trained for 1 epoch with a global training batch size of 256 and a PPO mini-batch size of 64. We use a learning rate of 5e-7 with 20 warmup steps for the actor model. To ensure stable optimization and manage memory efficiently, we freeze the vision tower, set the KL penalty coefficient to 0.001, and leverage Fully Sharded Data Parallel (FSDP) with both parameter and optimizer offloading.

8 Details of VST Data Generation

Table 8 presents the prompt templates utilized for our data generation pipeline. The entire data synthesis process strictly adheres to these formats to ensure high-quality, consistent annotations. As discussed in the Data Generation section, the LLM generates three distinct types of outputs: 1) video knowledge graph construction, which is conditioned on the current video segment and known entities to map dense physical relationships and object states (Tab. 8, top); 2) intermediate chain-of-thought (CoT) generation, which is conditioned on the specific time range and focused entities to provide incremental updates of the visual progress (Tab. 8, middle); and 3) multi-hop QA generation, which is conditioned on the constructed event reasoning path and visual evidence to synthesize practical reasoning question-answer pairs (Tab. 8, bottom).

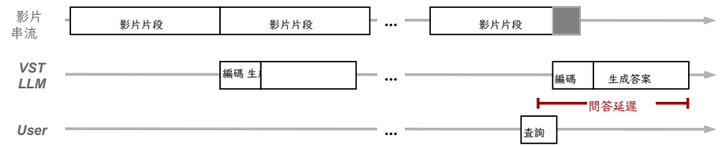


圖7：VST的串流推理流程。透過在使用者查詢到達前，為即將輸入的影片片段生成串流思考，VST能有效隱藏推理延遲，並實現快速問答響應。

7 VST-SFT / RL 訓練細節

VST-SFT超參數。為了訓練一個能夠持續思考的串流模型並確保可重現性，我們報告了VST-SFT階段的超參數。我們每個影片最多採樣384格，並限制最大影片像素為19,267,584，將影片token數上限設定為24K，以保留8K上下文給語言和推理。模型以學習率5e-6進行1個epoch訓練，並進行8次梯度累積步驟。

VST-RL 超參數。在強化學習階段，我們採用 GRPO 算法來優化模型。最大提示長度限制為 11,000 個 token，其中 1,000 個保留用於最大生成回應長度。在推演階段，我們使用 vLLM 以 1.0 的溫度和 0.98 的 top-p，為每個提示生成 8 個候選回應。模型以 256 的全域訓練批次大小和 64 的 PPO mini-batch 大小訓練 1 個週期。我們為行為模型使用 5e-7 的學習率，並進行 20 個 warmup 步驟。為了確保穩定優化和有效管理記憶體，我們凍結視覺塔，將 KL 懲罰係數設為 0.001，並利用 Fully Sharded Data Parallel (FSDP) 進行參數和優化器卸載。

8 VST 資料生成細節

表8呈現了我們數據生成管線所使用的提示模板。整個數據合成過程嚴格遵循這些格式，以確保高品質、一致的標註。如數據生成部分所述，LLM會生成三種不同的輸出類型：1) 影片知識圖構建，其基於當前影片片段和已知實體來映射密集的物理關係和物體狀態（表8，頂部）；2) 中間思維鏈（CoT）生成，其基於特定時間範圍和關注實體來提供視覺進度的逐步更新（表8，中間）；以及3) 多跳問答生成，其基於構建的事件推理路徑和視覺證據來綜合實用的推理問答對（表8，底部）。

Table 8: Templates of prompts used for data generation. The table presents three distinct prompts: Video Knowledge Graph Generation (top), Intermediate Chain-of-Thought (CoT) Generation (middle), and Question-Answer (QA) Generation (bottom). Curly-brace placeholders {...} will be replaced with actual segment-specific content during generation.

```
You are a Visual Scene Analyst specializing in dense scene graph generation. Your goal is to map ALL physical relationships in the video segment, not just human actions.
[CURRENT TIMELINE] Segment {#step_index}: {start_time}s to {end_time}s.
[CONTEXTUAL DATA] 1. Entity Registry: {known_entities_str}
[CRITICAL VISUAL EXTRACTION RULES] 1. NO "HUB-AND-SPOKE" BIAS (CRITICAL): Do not make the human protagonist the subject of every relation. Extract edges where neither the subject nor the object is a person. 2. Object-to-Object Relationships (MANDATORY): Look for Spatial Relations, Containment, and Passive Interactions. 3. Visual Entity Identification: Identify objects using specific visual descriptors. NO PRONOUNS. 4. Action & State Verbs: Use active verbs for humans and spatial/state verbs for objects. 5. Description: Describe the scene layout and object states, not just the human's movement.
[OUTPUT FORMAT] Return ONLY a JSON object: {"events": [{"subject": "...", "relation": "...", "object": "...", "description": "..."}]}
```

```
Current Video Segment: {current_time_range}
Task: Provide a incremental update of the visual scene/action/details in this segment. Focus on Progress (Delta) of visual elements.
Context & Entities: Focused Entities: {entity_text} (Integrate these naturally if they are actively involved in the current action. No full repetition each time; pronouns prohibited.)
Strict Constraints: 1. Focus on Dynamics: Describe WHAT is happening now with details. If the scene is static, be very concise. 2. NO restate: Do NOT restate the action/scene/details from the previous segment's end. Skip the action stated previously or state the changes. NEVER repeat. 3. Minimalist Reference: For entities already present/mentioned, use pronouns or minimal descriptors. 4. Language Variety: Avoid repetitive sentence structures. 5. No Redundancy: Do not repeat the timestamp or information from the previous context.
Output:
```

```
You are a Cognitive Video Intelligence Engine. Your task is to synthesize a Practical Deep Reasoning Question-Answer Pair based on the provided visual information from video segments and the Event Reasoning Path.
=== CORE DEFINITION (MUST FOLLOW FIRST) === 1. Event Reasoning Path Nature: The provided video segments are ordered by logical reasoning relevance, NOT by their original chronological time sequence. 2. Time Reference Rule: All time mentions must be based on the explicit time interval of each node. 3. Multi-hop Foundation: The reasoning must rely on logical connections between segments.
=== MANDATORY INSTRUCTIONS === 1. Strict Multi-hop Reasoning: The question MUST require integrating information from multiple video segments in the reasoning path. 2. Natural Language Constraint (CRITICAL): DO NOT use the words "Step", "Clip index", "Path node". Refer to segments using their time intervals or event descriptions. 3. Reasoning Dimensions: Construct the reasoning chain using one or more of the following logic types: {dimensions_str} 4. Practicality Requirement: The question must be meaningful for understanding the visual content of the video's narrative, intent, or physical logic.
=== OUTPUT FORMAT (JSON ONLY, NO EXTRA TEXT) === {"question": "...", "cot": "...", "answer": "...", "reasoning_type": "..."}
```


表8：用於數據生成的提示模板。該表展示了三種不同的提示：視頻知識圖生成（頂部）、中間因果推理（CoT）生成（中間）和問答（QA）生成（底部）。花括號占位符 {...} 將在生成過程中替換為實際的片段特定內容。

```
You are a Visual Scene Analyst specializing in dense scene graph generation. Your goal is to map ALL physical relationships in the video segment, not just human actions.
[CURRENT TIMELINE] Segment {#step_index}: {start_time}s to {end_time}s.
[CONTEXTUAL DATA] 1. Entity Registry: {known_entities_str}
    [關鍵視覺提取規則] 1。不允許「中心-輻射」偏差（關鍵）：不要讓人類主角成為每個關係的主語。提取主語和對象均非人的邊。2. 對象之間的關係（必須）：尋找空間關係、包含和被動互動。3. 視覺實體識別：使用特定的視覺描述符識別對象。不使用代詞。4. 動作與狀態動詞：使用主動...

verbs for humans and spatial/state verbs for objects  5  Description: Describe the scene layout and object states, not just the human's movement.
[OUTPUT FORMAT] Return ONLY a JSON object: {"events": [{"subject": "...", "relation": "...", "object": "...", "description": "..."}]}
```

```
Current Video Segment: {current_time_range}
Task: Provide a incremental update of the visual scene/action/details in this segment. Focus on Progress (Delta) of visual elements.
上下文與實體：聚焦實體：{entity_text}（如果它們是，則自然整合這些。
actively involved in the current action  No full repetition each time; pronouns prohibited.)
Strict Constraints: 1. Focus on Dynamics: Describe WHAT is happening now with details. If the scene is static, be very concise. 2. NO restate: Do NOT restate the action/scene/details from the previous segment's end. Skip the action stated previously or state the changes. NEVER
重複。3. 簡潔參考：對於已存在/提及的實體，使用代名詞或簡潔描述。4. 語言多樣性：避免重複的句子結構。5. 無。
Redundancy: Do not repeat the timestamp or information from the previous context
Output:
```

```
You are a Cognitive Video Intelligence Engine. Your task is to synthesize a Practical Deep Reasoning Question-Answer Pair based on the provided visual information from video segments and the Event Reasoning Path.
=== CORE DEFINITION (MUST FOLLOW FIRST) === 1. Event Reasoning Path Nature: The provided video segments are ordered by logical reasoning relevance, NOT by their original chronological time sequence. 2. Time Reference Rule: All time mentions must be based on the explicit time interval of each node. 3. Multi-hop Foundation: The reasoning must rely on logical connections between segments.
=== MANDATORY INSTRUCTIONS === 1. Strict Multi-hop Reasoning: The question MUST require integrating information from multiple video segments in the reasoning path. 2. Natural Language Constraint (CRITICAL): DO NOT use the words "Step", "Clip index", "Path node". Refer to segments using their time intervals or event descriptions. 3. Reasoning Dimensions: Construct the reasoning chain using one or more of the following logic types: {dimensions_str} 4. Practicality Requirement: The question must be meaningful for understanding the visual content of the video's narrative, intent, or physical logic.
=== OUTPUT FORMAT (JSON ONLY, NO EXTRA TEXT) === {"question": "...", "cot": "...", "answer": "...", "reasoning_type": "..."}
```